SUR 3103C – Geomatics

Course syllabus, Fall 2025

1 Course overview

Course description

This course primarily covers plane surveying techniques, including measurement of angles, distances, and elevation differences. Also covered are the related techniques of data reduction for these fundamental measurements, the Global Positioning System, topographic mapping, Earth-based coordinate systems, boundary surveys, and horizontal curves.

- 3 Credits
- Format: Varies by section
- https://elearning.ufl.edu/

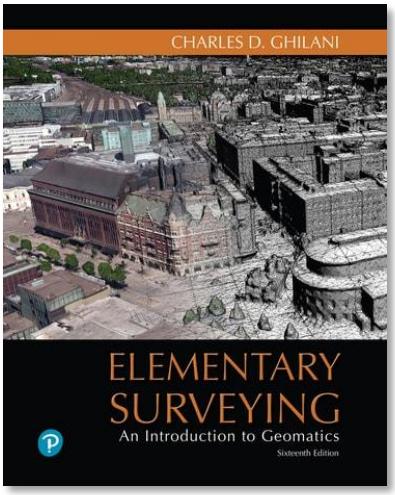
Prerequisites

MAC 1147 and MAC 2311 placement or equivalent.

Instructor

Andrew Lassiter, PhD

- **Contact**: Please email halassiter@ufl.edu from your UFL email address. **Begin the subject line with SUR 3103C** to help me find it. (I get so many emails.)
- **Office location & hours**: 8th period Wednesday and Thursday (3–4 PM) or by appointment.
    - In-person: Reed Laboratory 406A
    - Zoom: See Canvas for link



Text

Ghilani, C. *Elementary Surveying: An Introduction to Geomatics*. 16th Edition, Pearson Education, 2021.
(Older editions 12th–15th are probably okay.)

2 Learning outcomes

By the end of this course, students must demonstrate knowledge and ability in the following:

- Significant figures, accuracy, and precision
- Types and sources of error
- Measurement of accurate horizontal distances, and horizontal and vertical angles
- Computation of horizontal coordinates by traverse adjustment
- Making vertical measurements by differential leveling
- Computation of elevations by level loop adjustment

- Bearing and azimuth calculation
- Computation of area of a parcel of land
- Production of a large-scale topographic map

By the end of this course, students should have gained general (introductory) knowledge in:

- Coordinate systems and datums
- The Global Positioning System
- Boundary surveys
- The United States Public Land Survey System

3 Course logistics

This course is delivered in five modules, each about three weeks long. A typical module consists of:

- **Two live lectures per week** to be attended in-person by Gainesville sections
    - These lectures are broadcast live via Zoom for students at UF satellite campuses and certificate students.
    - The recordings of these lectures will be available in the evening after the lectures. The recordings are not a substitute for attendance.
- A **weekly field or office lab exercise**. (Some lab exercises span two weeks.)
- A **homework assignment** based on the contents of the module.
    - There is a homework assignment to begin the semester: The **trigonometry review**.
- A **module quiz** based on the module's contents.
    - The final module will not have its own quiz; its contents will be covered in the **comprehensive final exam**.

Class meeting times

Tuesday and Thursday 7th period (1:55 PM – 2:45 PM Eastern Time) in Reed Laboratory 302

See Canvas for live Zoom link.

- **Gainesville students:** You are divided into two lab sections. One meets Monday 8th – 10th period (3:00 PM – 6:00 PM) and the other on Tuesday at that same time.
- **UF satellite students**: Your rec center coordinator will contact you with lab meeting times.
- **Certificate students**: Your lab schedule is flexible to meet your schedule. Any lab due dates you see in this syllabus or on Canvas are only suggestions to keep you on track.

Technology requirements

- A computer or mobile device with high-speed internet connection.
- A webcam, headset and/or microphone, and speakers.
- Latest version of Web browser. Canvas supports only the two most recent versions of any given browser.
- Installation of proctoring software may be required and will be provided if so.

3.1 Description of assessments & activities

Certificate students: Some of your point allocations and descriptions below will differ slightly.

Introduction

Introduce yourself in the Canvas discussion set up for that purpose. Read and comment on your peers' introductions. This is a great way to meet students at other campuses.

Quizzes

Quizzes will be administered once per module through Canvas (see schedule). Quizzes will consist of multiple choice, short answer, and calculations. Quizzes will open and close on the dates noted.

Quizzes are closed book and closed notes and are proctored. More details on proctoring are available on Canvas.

Homework

Homework assignments are typically focused on calculations and will vary between hand-completed documents and Canvas quiz-style assignments. Detailed instructions are given for each assignment.

Lab exercises

Students will be set into groups of two or three to complete lab assignments both in the field and in the office that pertain to the survey of a small parcel. These assignments all tie together and culminate in the final project, the topographic mapping project.

Topographic mapping project

The final product of the semester's lab exercises is a topographic map of your parcel, which is completed in AutoCAD.

Lab attendance

Any absence from a lab period must be coordinated with your lab group to ensure the work is completed, but no credit for attendance can be made up.

Field book and lettering exercise

Data and procedures in the field are recorded by hand in a field book. One field book is kept per group. Except for the **Station Descriptions** lab, notetaking should be performed by one group member per field lab exercise. At the end of the semester, each group member will submit a scan of the complete field book and note their own best notetaking example from the semester.

Part of the field book grade is the **Lettering** exercise, where each student practices their lettering at the beginning of the semester and once again at the end of the semester to demonstrate improvement.

Comprehensive final exam

The final exam is cumulative, covering the entire semester's topics. The proctoring rules mentioned above apply.

Fall 2025 final exam date and time: Thursday, December 11, 2025, 3:00 PM – 5:00 PM
Exceptions to this date and time for non-degree-seeking students are available on a case-by-case basis.

Attendance and make-up work

Lab attendance is mandatory. Lecture attendance is strongly encouraged for Gainesville students and those students who can feasibly attend live. There is a strong correlation between attendance and success in college courses.

Requirements for class attendance and make-up exams, assignments and other work are consistent with university policies that can be found at:

https://catalog.ufl.edu/UGRD/academic-regulations/attendance-policies/

3.2 Grades & grading scale

Item	Points
Introduction discussion	20
Quizzes: 4 quizzes @ 30 points each	120
Homework: 6 assignments @ 15 pts each	90
Lab exercises: 10 exercises @ 25 pts each	250
Lab attendance: 12 sessions @ 5 pts each	60
Topographic mapping project	60
Field book & lettering	20
Comprehensive final exam	150
TOTAL POSSIBLE POINTS	**770**

***Certificate students**: Your mentor form and parcel plan are not graded, but you will not have access to lab assignments until they are submitted and accepted.*

Grading scale (%)

A	[0.94, ∞)
A-	[0.90, 0.94)
B+	[0.87, 0.90)
B	[0.84, 0.87)
B-	[0.80, 0.84)
C+	[0.77, 0.80)
C	[0.74, 0.77)
C-	[0.70, 0.74)
D+	[0.67, 0.70)
D	[0.64, 0.67)
D-	[0.60, 0.64)
E	(−∞, 0.60)

Grades and grade points

For information on current UF policies for assigning grade points, see:

https://catalog.ufl.edu/UGRD/academic-regulations/grades-grading-policies/

4 Learning content

4.1 Lectures and reading list

(1) Course introduction. Written field notes. **Read**: Ch. 1 (all) Ch. 2 § 6–11.
(2) Units and significant figures. **Read**: Ch. 2 § 1–5.
(3) Theory of errors in observations. **Read**: Ch. 3 § 1–8; **Browse**: Ch. 3 § 9–21.
(4) Distance measurement. **Read**: Ch. 6 § 1–13.
(5) Differential leveling. **Read**: Ch. 4 (all).
(6) Leveling procedures and computations. **Read**: Ch. 5 (all).
(7) Taping corrections. **Read**: Ch. 6 § 14, Appendix A.1.
(8) Electronic distance measurement. **Read**: Ch. 6 § 15–23.
(9) Angles, azimuths, and bearings. **Read**: Ch. 7 § 1–9.
(10) Compass readings. Total Stations. **Read**: Ch. 7 § 10–16; Ch. 8 § 1–6.
(11) Horizontal and vertical angle measurement. **Read**: Ch. 8 § 7–22.
(12) Traversing. **Read**: Ch. 9 (all).
(13) Traverse computations. **Read**: Ch. 10 § 1–6.
(14) & (15) Traverse adjustment. **Read**: Ch. 10 § 7–17.
(16) Area: Coordinate and DMD methods. **Read**: Ch. 12 (all).
(17) & (18) Global navigation satellite systems (GNSS) intro and principles. **Browse**: Ch. 13 (all).
(19) GNSS operations. **Read**: Ch. 14 § 1–2; **Browse**: Ch. 14 § 3–7; Ch. 15.
(20) Data collectors. **Read**: Ch. 2 § 12–15.
(21) Mapping surveys. **Read**: Ch. 17 § 1–10, 12–16.
(22) Interpreting and drawing contours. **Review**: Ch. 17 § 5–8, 9.2, 9.3.
(23) Introduction to Mapping and Cartography. **Read**: Ch. 18 (all).
(24) Control surveys and Geodetic datums. **Read**: Ch. 19 § 1–12; **Browse**: Ch. 19 § 13–14.
(25) State plane coordinates. **Read**: Ch. 20 § 1–5, 8–8.1, 9–11; **Browse**: rest of chapter.
(26) Boundary surveys. **Read**: Ch. 21 (all).
(27) & (28) United States Public Land Survey System. **Read**: Ch. 22 (all).
(29) *(if time allows)* Horizontal curves. **Read**: Ch. 24 § 1–4.

4.2 Lab exercises

(1) Station descriptions; Lettering pt. I
(2) Distances – taping and pacing
(3) Differential Leveling
(4) Horizontal distances
(5) & (6) Horizontal angle measurement
(7) Traverse adjustment
(8) GNSS exercise
(9) CAD exercise
(10) Planimetric mapping
(11) Contour mapping
(12) Map drafting; Lettering pt. II

5 Policies and requirements

This course plan and syllabus are subject to change in response to student and instructor needs. Any changes will be clearly communicated in advance through Canvas.

CALS and UF Policies can be reviewed at:

https://syllabus.ufl.edu/syllabus-policy/uf-syllabus-policy-links/

5.1 Communication courtesy and professionalism

Just as in any professional environment, meaningful and constructive dialogue is expected in this class and requires a degree of mutual respect, willingness to listen, and tolerance of opposing points of view. **Respect for individual differences and alternative viewpoints will be maintained in this class at all times.** All members of the class are expected to follow rules of common courtesy, decency, and civility in all interactions. Failure to do so will not be tolerated and may result in loss of participation points and/or referral to the Dean of Students' Office.

5.2 Privacy statement

Our class sessions may be audio-visually recorded for students in the class to reference and for enrolled students who are unable to attend live. Students who participate with their camera engaged or utilize a profile image agree to have their video or image recorded. If you are unwilling to consent to have your profile or video image recorded, be sure to keep your camera off and do not use a profile image. Likewise, students who un-mute during class and participate orally are agreeing to have their voices recorded. If you are not willing to consent to have your voice recorded during class, you will need to keep your mute button activated and communicate exclusively using the "chat" feature, which allows students to type questions and comments live. The chat will not be recorded or shared. As in all courses, unauthorized recording and unauthorized sharing of recorded materials is prohibited.

5.3 Software use

All faculty, staff and students of the university are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against university policies and rules, disciplinary action will be taken as appropriate.

6 Calendar

This calendar will be followed as closely as possible but is subject to change.

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
8/18	**8/19**	**8/20**	**8/21** Lec. 1	**8/22**	**8/23**
8/25 Lab 1	**8/26** Lec. 2 Lab 1	**8/27**	**8/28** Lec. 3	**8/29**	**8/30**
9/1 **Labor Day** *No lab*	**9/2** Lec. 4 *No lab*	**9/3**	**9/4** Lec. 5	**9/5**	**9/6** **REC Labs 1 & 2**
9/8 **Trig review due** Lab 2	**9/9** Lec. 6 Lab 2	**9/10**	**9/11** Lec. 7	**9/12**	**9/13**
9/15 **HW #1 due** Lab 3	**9/16** Lec. 8 Lab 3	**9/17**	**9/18** Lec. 9 **Quiz 1**	**9/19**	**9/20**
9/22 Lab 4	**9/23** Lec. 10 Lab 4	**9/24**	**9/25** Lec. 11	**9/26**	**9/27** **REC Labs 3 & 4**
9/29 Lab 5/6	**9/30** Lec. 12 Lab 5/6	**10/1**	**10/2** Lec. 13	**10/3**	**10/4** **REC Lab 5/6**
10/6 **HW #2 due** Lab 5/6	10/7 Lec. 14 Lab 5/6	10/8	10/9 Lec. 15 **Quiz 2**	10/10	10/11 **REC Lab 7 (Excel)**
10/13 Lab 7	**10/14** Lec. 16 Lab 7	**10/15**	**10/16** Lec. 17	**10/17** **Homecoming**	**10/18**
10/20 Lab 8	**10/21** Lec. 18 Lab 8	**10/22**	**10/23** Lec. 19	**10/24**	**10/25** **REC Lab 8**
10/27 **HW #3 due** Lab 9	**10/28** Lec. 20 Lab 9	**10/29**	**10/30** Lec. 21 **Quiz 3**	**10/31**	**11/1** **REC Lab 9 (CAD)**
11/3 Lab 10	**11/4** Lec. 22 Lab 10	**11/5**	**11/6** Lec. 23	**11/7**	**11/8**
11/10 **Veterans Day** *No lab*	11/11 Lec. 24 *No lab*	11/12	11/13 Lec. 25	11/14	11/15 **REC Labs 10 & 11**
11/17 Lab 11	**11/18** Lec. 26 Lab 11	**11/19**	**11/20** Lec. 27 **Quiz 4**	**11/21** **HW #4 due**	**11/22** **REC Lab 12**
11/24 **Thanksgiving** *(all week)*	**11/25**	**11/26**	**11/27**	**11/28**	**11/29**
12/1 Lab 12	**12/2** Lec. 28 (& 29?) Lab 12	**12/3**	**12/4** **Reading day**	**12/5** **Reading day**	**12/6**
12/8 **HW #5 due**	**12/9**	**12/10**	**12/11** **Final exam** 3:00 PM – 5:00 PM	**12/12**	**12/13**